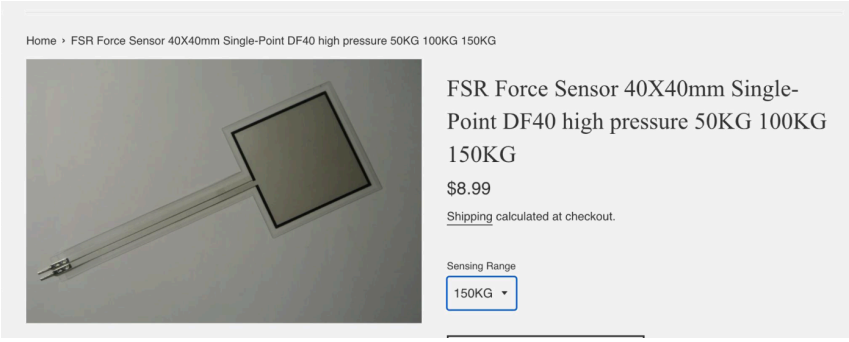
Prototyping 3

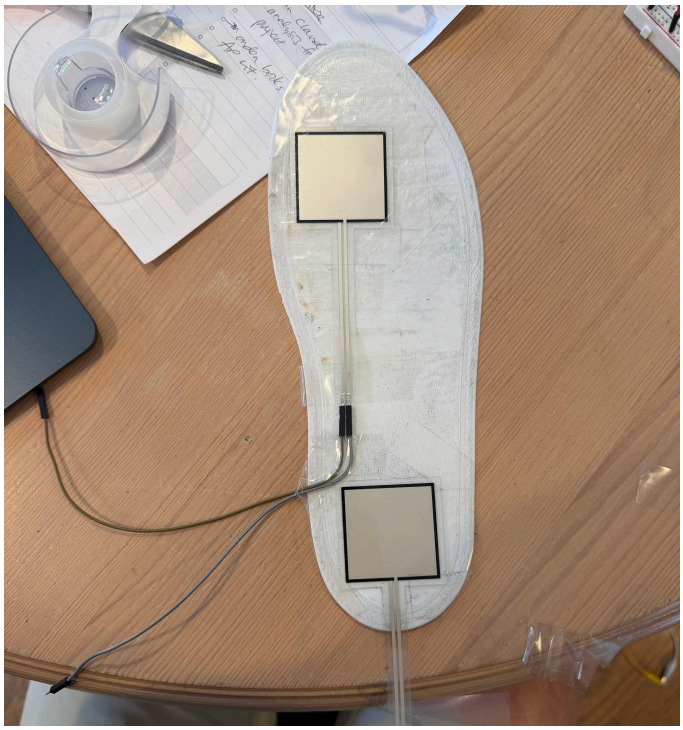
Michael Phelan, Concordia Lutheran High School | Dr. T. Jenkyn, Western University |

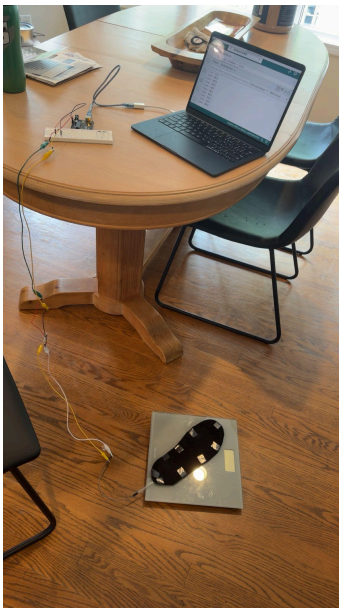
For this prototyping phase, we got a lot done. To start it off, I decided to buy these FSRs from a site called yanlania. It is because they were the cheapest FSRs I could find that could hold up to 150kg of weight.



When they arrived, I tested them right away. I hooked the sensors up and did a very basic test. I just pressed as hard as I could and saw that there was a really wide range of points it covered. So, I concluded that it could measure body weight.

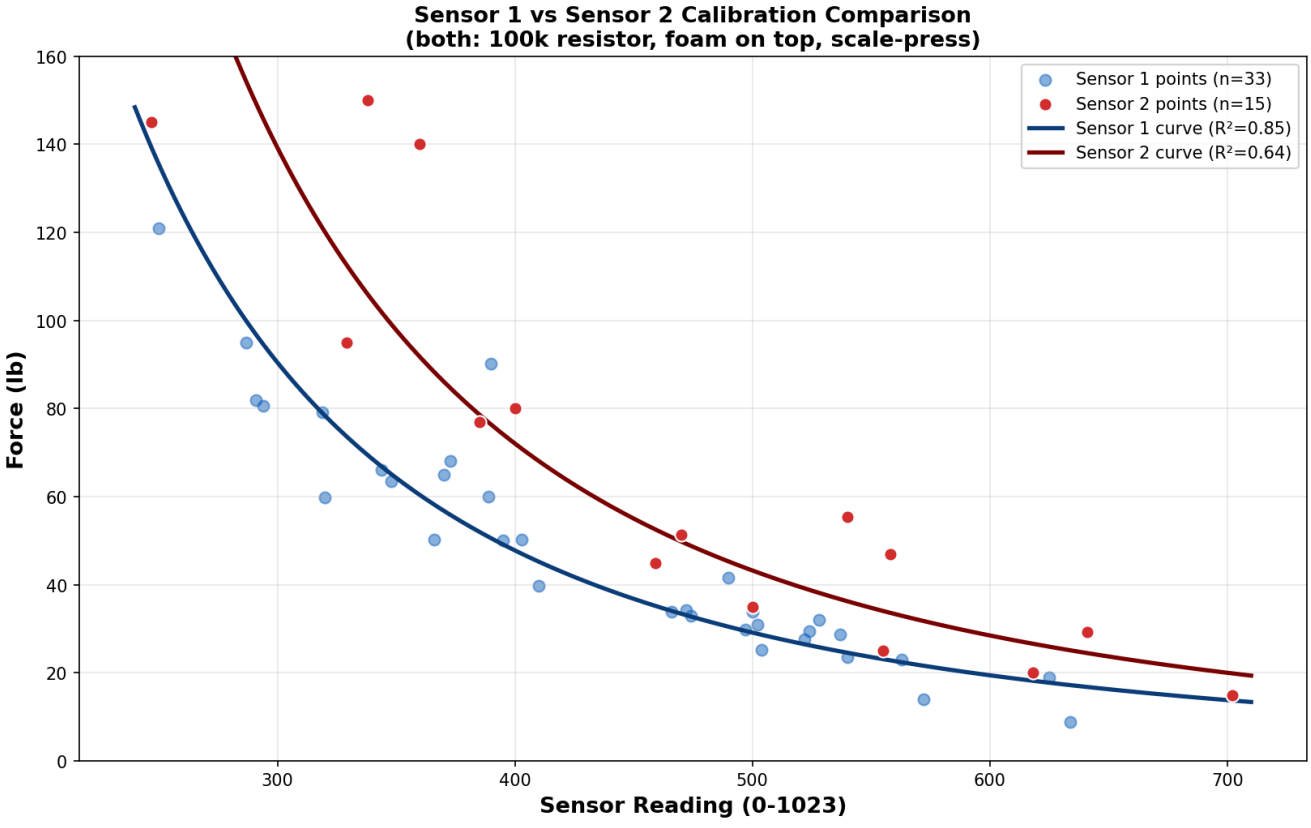
I got right into testing. As talked about before, I taped one sensor on the ball and one sensor on the heel on my 3d printed insole. I then covered this up with foam.



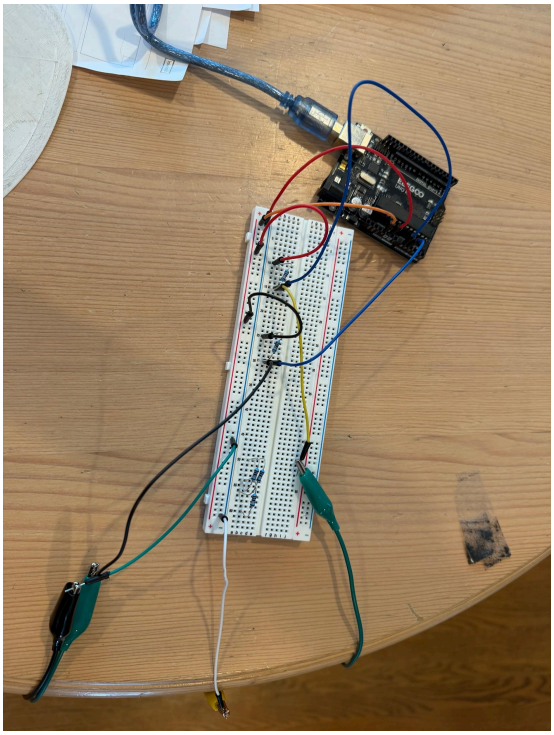


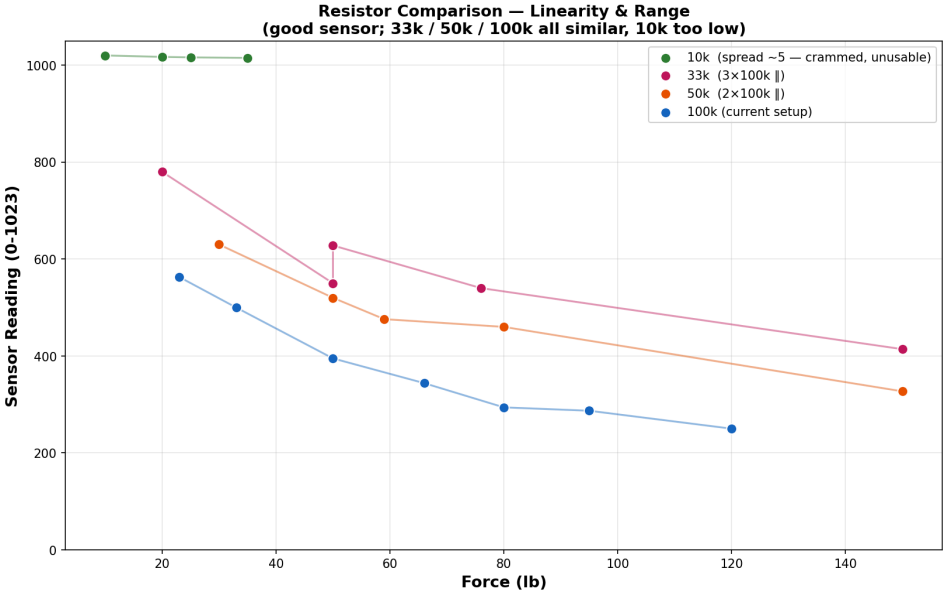
I then calibrated the sensors. To do this, I got a bathroom scale, I stepped on the bathroom scale until it locked in a certain weight, then looked at the readings on the screen. I repeated this process until I had over 30 points. This is just a really rough calibration, so I will need to calibrate it better in the lab.

For both sensors, I plotted the known weight I pressed down, and the adjacent reading value on screen. The graph is below. Overall, they both had pretty shaky calibration curves, with sensor one (the heel sensor) having an R^2 of 0.85, and sensor two (the ball sensor) having an R^2 of 0.64



I then combined both sensors on the same breadboard so both readings could be read at the same time. The code was changed so it would read the readings, then using the calibration curves to convert the readings to weight. However, when the insole is in the shoe, there is pressure applied by your foot. So I added a zero and max feature. When you start up the program. It will ask you to lift your foot up and put zero percent of your weight on it. Then it will ask for 100% to get a range. The readings will then fall anywhere in the range. However, it is not capped at 100%. For example, sometimes if you jump it will spike.





I also tried out different resistors to fix linearity, but they were basically the same. So I stuck with 100k



After calibrating, I tried a full shoe test. I put the insole in my shoe and tried it on. The zeroing helped a lot. And I honestly think it was pretty accurate.

To test how accurate the device was, I used a two-scale method. I stood with my left foot on a bathroom scale and my right foot (the one with the sensor insole) on the ground. Since my total body weight is 177 lb, whatever the scale reads under my left foot tells me how much weight is on my right foot: the right foot carries 177lb minus the scale reading. For example, if the scale reads 20 lb, then my right foot is bearing 177 − 20 = 157 lb. By shifting my weight between my two feet, I could control how much load went onto the sensor and check it against a known value. I tested several points across the range:

- Scale read 88 lb → right foot bearing 89 lb (50% of body weight). Device showed ~40-45%.
- Scale read 120 lb → right foot bearing 57 lb (32%). Device showed ~30%.
- Scale read 50 lb → right foot bearing 127 lb (72%). Device showed ~60%.

Across these points, the device was consistently within about 10-15% of the true value, and always read in the correct direction (more weight gave a higher reading). For a device built from two low-cost FSRs and calibrated by hand, this is a reasonable level of accuracy. It is not precise enough for strict clinical use, but it works well as a rough real-time guide for partial weight-bearing, which was the goal. The device also correctly read 0% when the foot was fully unloaded and roughly 100% at full body weight.

Side note: this was taken from only 3 data points. That was a very rough conclusion. Again, to fully test the validity of the device, I need to test it in the lab.

I then did a walking test. I changed the code so that I could stop the live reading, and then hit "g" to record five seconds of data. I did four walking tests, carrying everything in my hands: all the wires, my computer, and the breadboard. For each one, I hit "g," walked for five seconds across my house, and repeated that four times.

The device was consistent in showing the walking pattern: each step clearly registered as my weight going on and off the foot. However, it was also very consistent in showing a spike to around 200% body weight each time my foot hit the ground.

I did some research into this. The research shows that when your foot hits the ground during normal walking, the force typically peaks at around 115-125% of body weight, not 200%. So there's a discrepancy here that we need to figure out. The device is reading the foot-strike impact much higher than it should be. I think that this comes from calibrating the device statically (while standing still), which doesn't accurately capture the higher, dynamic forces that happen during an actual step.

Overall, the device shows a general walking curve; very general, but the step pattern is clearly there and repeatable across all four trials.

